

CO4-AT2-PART

DESIGN AND DEVELOP MODEL – SAMPLING-BASED PLANNING & DATA GENERATION

REINFORCEMENT LEARNING-BASED ROBOT NAVIGATION USING SAMPLING-BASED PLANNING AND SYNTHETIC DATA GENERATION

1. ABSTRACT

Autonomous robot navigation requires an efficient method for reaching a target while avoiding obstacles. This project presents a navigation model combining synthetic data generation, sampling-based planning, and reinforcement learning. The Rapidly-exploring Random Tree Star (RRT*) algorithm is used for sampling-based path planning, while Proximal Policy Optimization (PPO) is used to train an agent to learn navigation decisions.

2. INTRODUCTION

Robot navigation is an important problem in autonomous robotics. A robot needs to move from a starting position to a desired goal while avoiding obstacles. In complex environments, finding a suitable path can be computationally challenging.

Sampling-based planning methods solve this problem by randomly exploring the configuration space. **RRT** creates a rapidly expanding search tree, while **RRT*** improves the generated solution by selecting lower-cost paths and rewiring the tree.

2.1 Problem Statement

To design and develop an intelligent robot navigation model that can operate in randomly generated obstacle environments by combining sampling-based planning with reinforcement learning.

2.2 Objectives

- Generate synthetic navigation environments.
- Generate random obstacles, start and goal positions.
- Implement RRT* for collision-free path planning.
- Develop an RL-based navigation environment.
- Train a PPO agent.
- Evaluate navigation performance using suitable metrics.

3. PROPOSED SYSTEM

The proposed system contains four major modules.

Module 1 – Synthetic Data Generation

Random two-dimensional environments are created with different obstacle configurations. Each scenario contains a robot starting position, a goal position, and randomly generated obstacles.

Module 2 – RRT* Sampling-Based Planning

RRT* randomly samples points from the environment and constructs a search tree. Collision checking prevents the path from passing through obstacles. Rewiring is used to improve the path cost.

Module 3 – PPO Reinforcement Learning

The navigation problem is formulated as an RL environment. The robot can move in four directions: up, down, left, and right. PPO learns a policy by maximizing the accumulated reward.

Module 4 – Performance Evaluation

The trained model is evaluated using:

- Success rate
- Average reward
- Number of steps
- Path length

4. METHODOLOGY AND IMPLEMENTATION

4.1 Synthetic Data Generation

A two-dimensional environment of size 20×20 is used. Random rectangular obstacles are generated to create different navigation scenarios.

The following short code generates 1000 synthetic records:

```
import random
import pandas as pd
def generate_obstacles(n=8):
    return [(random.randint(2,16),
              random.randint(2,16),
              random.randint(1,3),
              random.randint(1,3)) for _ in range(n)]
data = []
df.to_csv("synthetic_navigation_data.csv", index=False)
print(df.head())
```

The generated dataset contains **1000 synthetic navigation scenarios**.

4.2 Collision Detection

Collision detection is necessary to ensure that the robot does not move through an obstacle.

If the robot position lies inside an obstacle, the function returns True.

4.3 RRT* Sampling-Based Planning

RRT* begins from the starting point and randomly samples positions in the environment. The nearest tree node is selected and a new node is generated toward the sampled point.

A simplified implementation is:

```
def rrt_star(start, goal, obstacles, iterations=800):  
    nodes = [Node(start[0], start[1])]   
    nearest = min(  
        nodes,  
        key=lambda n:(n.x-sample.x)**2+(n.y-sample.y)**2  
    )  
    return None
```

The algorithm generates a path by repeatedly sampling the environment and expanding the tree toward feasible locations.

5. REINFORCEMENT LEARNING MODEL

5.1 State Space

The state represents the robot's current position and the target position.

The normalized state is:

State = [Robot X, Robot Y, Goal X, Goal Y]

Normalization keeps the values between 0 and 1.

5.2 Action Space

The robot has four possible actions:

Action	Movement
0	Up
1	Down
2	Left
3	Right

5.3 Reward Function

The reward function encourages the robot to reach the goal efficiently.

- **+100** → Goal reached
- **-10** → Collision
- **-1** → Normal movement
- **+1** → Movement closer to the goal

10. CONCLUSION

This project developed an intelligent robot navigation framework using synthetic data generation, RRT* sampling-based planning, and PPO reinforcement learning. Synthetic environments containing random obstacles were generated automatically. RRT* was used to demonstrate sampling-based path planning, while PPO learned navigation decisions through interaction with the simulated environment.

The model provides a practical demonstration of how traditional path-planning techniques can be combined with reinforcement learning. The generated paths, training rewards, success rate, and navigation steps provide useful measures for evaluating the developed system.

The project successfully demonstrates the design and development of a model based on sampling-based planning and data generation, making it suitable for reinforcement learning assessment and further extension to real-world robotic navigation.

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